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Convolution theorems for the free metaplectic transformation and its application

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Abstract

The free metaplectic transformation (FMT) is widely used in several fields, including filter design, pattern recognition, image processing, and optics. This study investigated two new convolution theorems in the FMT domain to obtain a more concise and intuitive convolution form. First, on the basis of expressing the generalized translation, we derived the convolution theorem (of the first kind) in the FMT domain, which has the elegance and simplicity of the classical Fourier transform (FT) results. Second, we provide the convolution theorem (of the second kind) in the FMT domain to further study the diversity of the convolution theory. The provided solution can be represented using a simple integral that is easy to implement in multiplying filter designs. Finally, based on this simple form of the convolution theorems, we designed a multiplicative filter in the FMT domain, demonstrating the feasibility and efficiency of the filters designed in this study through simulations.

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1. Introduction

The concept of the metaplectic group was proposed by Segal [1] and Shale [2], and various theoretical studies related to this concept were developed in mathematics and physics [3–10]. In 1989, Howe et al. [11] discussed the fundamental role of metaplectic representation in the mathematical analysis field. In particular, the metaplectic representation has attracted the

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attention of various researchers [12–17], who have proposed this representation under various special forms, such as quadratic Fourier transform [12,13], linear canonical transform (LCT) [18–20], Fresnel transform (FRT) [21], fractional Fourier transform (FRFT) [7,22,23]. The free metaplectic transformation (FMT) was first proposed in [24] as a special form of metaplectic representation. In essence, FMT [25–28] is similar to the well-known LCT; therefore, it is also called an N -dimensional (N-D) non-separable LCT. However, unlike conventional N-D LCT, FMT is generated using a general $2N \times 2N$ free symplectic matrix $\mathbf{M}$ with $N(2N + 1)$ degrees of freedom [29]. Owing to these additional degrees of freedom, FMT is widely used in fields such as filter design, pattern recognition, image processing, quantum mechanics, optics, and electromagnetic wave propagation analysis [30–36].

Note that convolution theory plays a crucial role in mathematics, signals, images, and optics, particularly in designing and implementing multiplicative filters [37–45]. In recent years, many generalizations of the classical convolution theory have been developed to extend this theory to a wider field, where a large number of convolution theorems on LCT have been obtained [18,19,46–50]. These solutions have dealt with one-dimensional (1-D) signal problems. Although some researchers are currently studying related theories in the multi-dimensional signal space—such as LCT convolution and sampling, their research results are not yet mature. Recent findings aimed to exploit the advantages of LCT in analyzing multidimensional signals [51–54]. However, most existing multidimensional LCT studies have been conducted in kernels, which are essentially the product of N copies of a typically 1-D LCT kernel. These studies do not consider the inherent multidimensional nature of LCT. In this regard, considering the pure and complex multidimensional structure of LCTs, we initiate a special study in the field of multidimensional signals called FMT utilizing a pure multidimensional kernel derived from a general $2N \times 2N$ real symplectic matrix $\mathbf{M}$. Currently, theories such as convolution, sampling, and the uncertainty principle are still at the development stage in the FMT field. Consequently, studying the convolution theorem in the field of FMT is of great theoretical relevancy.

In addition, the convolutional theory of FMT is an interesting topic that requires more research in this area. Moreover, the convolution theorem of the classical Fourier transform (FT) shows that the traditional convolution of two signals is equivalent to the product of FT of these two signals. Although a new convolution theorem in the FMT domain has been proposed based on the N-D FT definition, in terms of the form of the convolution theorem, this new theorem is more complicated than the classical FT domain convolution theorem, and it is difficult to apply [33]. Shah generalized FT classical convolution to the FMT domain and derived convolution theorem containing additional chip multiplier terms [34]. To obtain a more concise and intuitive convolution theory, we used the transform kernel of FMT to propose two types of convolution structures, the results of which are as elegant and simple as possible compared with the FT. The advantage of these two convolution structures is that they are simple in form, easy to implement, and are not affected by any chip multiplier term.

The contribution of this study can be divided into four parts:

- (1) The novelty of this study is that a pure multidimensional kernel is derived from a real symplectic matrix $\mathbf{M}$ of a general $2N \times 2N$ to study convolution theory in the field of multidimensional signals, while most of the existing multidimensional signal researches adopt the form of a transform kernel, which are essentially the product of N copies of the usual 1-D transform kernel.
- (2) Based on the expression of the generalized translation, we derive the convolution theorem (of the first kind) in the FMT domain. The advantage is that it has elegance and

simplicity comparable to classical FT results. The disadvantage is that the generalized convolution expression is in the form of a triple integral, making it difficult to convert it to a single integral form.

- (3) Due to the complexity of the triple integrals, we propose a convolution theorem (of the second kind) in the FMT domain. The proposed theorem is simple integral and easy to implement in multiplicative-filter designs. However, it cannot quite maintain the elegance and simplicity comparable to classical FT results by adding a contraction factor of $\frac{1}{\sqrt{2}}$ in the FMT domain.
- (4) The two types of convolution theorems proposed in this paper are expressed in the form of a simple product of two FMTs in the FMT domain and are unaffected by any chip-multiplier terms, even though the result of the convolution theorem of the second type has a contraction factor of $\frac{1}{\sqrt{2}}$. Based on the simplicity of this result, these theorems are easy to implement in filter design.

The remainder of this paper is organized as follows. [Section 2](#) discusses the basics of FMT and the convolution theory. In [Section 3](#), two types of convolution theorems in the FMT domain are presented. In [Section 4](#), we present a multiplicative-filter design in the FMT domain based on these two types of convolution theorems. Finally, [Section 5](#) concludes the study.

2. Preliminaries

In this section, we review the basics of FMT, generalized translation, and the general framework of convolution theory, which are discussed throughout the paper.

2.1. Free metaplectic transformation

First, some background on the FMT is essential. The FMT of a signal $f(\mathbf{t})$ with a symplectic matrix $\mathbf{M} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, where $\det(B) \neq 0$, can be defined as follows [26,32]:

$$\mathcal{L}_{\mathbf{M}}[f(\mathbf{t})](\mathbf{u}) = \widehat{f}_{\mathbf{M}}(\mathbf{u}) = \int_{\mathbb{R}^N} f(\mathbf{t}) \mathcal{K}_{\mathbf{M}}(\mathbf{t}, \mathbf{u}) d\mathbf{t}, \quad (1)$$

where the kernel function $\mathcal{K}_{\mathbf{M}}(\mathbf{t}, \mathbf{u})$ can be expressed as follows:

$$\mathcal{K}_{\mathbf{M}}(\mathbf{t}, \mathbf{u}) = \frac{1}{|\det(B)|^{\frac{1}{2}}} e^{\pi i (\mathbf{u} D B^{-1} \mathbf{u}^T + \mathbf{t} B^{-1} A \mathbf{t}^T) - 2\pi i \mathbf{t} B^{-1} \mathbf{u}^T}, \quad (2)$$

where $\mathbf{u} = (u_1, u_2, \dots, u_N)$, $\mathbf{t} = (t_1, t_2, \dots, t_N)$, $A = (a_{kl})$, $B = (b_{kl})$, $C = (c_{kl})$, and $D = (d_{kl})$ are all $N \times N$ real matrices that satisfy the following constraints [12,13,26,32]:

$$AB^T = BA^T, \quad CD^T = DC^T, \quad AD^T - BC^T = I_N,$$

where I_N denotes an $N \times N$ identity matrix.

The corresponding inverse formula can be expressed as follows:

$$\begin{aligned} f(\mathbf{t}) &= \mathcal{L}_{\mathbf{M}^{-1}}[\widehat{f}_{\mathbf{M}}(\mathbf{u})](\mathbf{t}) \\ &= \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} \widehat{f}_{\mathbf{M}}(\mathbf{u}) e^{-\pi i (\mathbf{u} B^{-T} D^T \mathbf{u}^T + \mathbf{t} A^T B^{-T} \mathbf{t}^T) + 2\pi i \mathbf{u} B^{-T} \mathbf{t}^T} d\mathbf{u}, \end{aligned} \quad (3)$$

Table 1

Well-known integral transformations for specific combinations of blocks found in the symplectic matrix.

Free metaplectic matrix $\mathbf{M} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$	Free metaplectic transformation
$A = D = \mathbf{0}, B = -C = I_N$	N-D FT
$A = \text{diag}(a_{11}, \dots, a_{NN}), B = \text{diag}(b_{11}, \dots, b_{NN}),$ $C = \text{diag}(c_{11}, \dots, c_{NN}), D = \text{diag}(d_{11}, \dots, d_{NN})$	N-D separable LCT
$A = D = \text{diag}(\cos \alpha_1, \dots, \cos \alpha_N),$ $B = -C = \text{diag}(\sin \alpha_1, \dots, \sin \alpha_N)$	N-D separable FRFT
$A = D = I_N \cos \alpha, B = C = I_N \sin \alpha$	N-D non-separable FRFT
$A = D = I_N, B = \text{diag}(b_{11}, \dots, b_{NN}), C = \mathbf{0}$	N-D separable FRT
$A = D = I_N, C = \mathbf{0}$	N-D non-separable FRT

where $\mathbf{M}^{-1} = \begin{pmatrix} D^T & -B^T \\ -C^T & A^T \end{pmatrix}$.

The FMT has $N(2N + 1)$ degrees of freedom [29], which can be reduced to some well-known integral transformations for specific combinations of blocks found in the symplectic matrix, such as N-D FT, N-D LCT, N-D FRFT, and N-D FRT, which are summarized in Table 1.

2.2. Generalised translation and general framework of convolution theory

In the general framework of convolution theory [55,56], most of the existing convolution theories of integral transforms are not the most concise and intuitive compared with the convolution theory of the classical FT; consequently, general convolution and related theories have been proposed [19,46–49,57]. Let us review the related content.

If we consider a general signal transform and its Fourier-type inverse, expressed as follows:

$$f(\mathbf{t}) = \int_{\mathbb{R}^N} \rho(\mathbf{u}) F(\mathbf{u}) \mathcal{K}(\mathbf{u}, \mathbf{t}) d\mathbf{u}, \quad (4)$$

$$F(\mathbf{u}) = \int_{\mathbb{R}^N} \rho(\mathbf{t}) f(\mathbf{t}) \mathcal{K}^*(\mathbf{u}, \mathbf{t}) d\mathbf{t}. \quad (5)$$

Then, the $\boldsymbol{\tau}$ -generalised translation of $f(\mathbf{t})$ denoted by $f(\mathbf{t}\boldsymbol{\theta}\boldsymbol{\tau})$ can be expressed as follows [42]:

$$f(\mathbf{t}\boldsymbol{\theta}\boldsymbol{\tau}) = \int_{\mathbb{R}^N} \rho(\mathbf{u}) F(\mathbf{u}) \mathcal{K}(\mathbf{u}, \mathbf{t}) \mathcal{K}^*(\mathbf{u}, \boldsymbol{\tau}) d\mathbf{u}, \quad (6)$$

where $\boldsymbol{\theta}$ in the argument of the function $f(\mathbf{t}\boldsymbol{\theta}\boldsymbol{\tau})$ denotes the generalised delay operator for the generalised translation, $\mathcal{K}(\mathbf{u}, \mathbf{t})$ denotes the kernel of the transform, $\mathcal{K}^*(\mathbf{u}, \boldsymbol{\tau})$ denotes the complex conjugate of $\mathcal{K}(\mathbf{u}, \boldsymbol{\tau})$, and $\rho(\mathbf{u})$ denotes the weight function.

The shift property for N-D FT can be expressed as follows:

$$f(\mathbf{t} - \boldsymbol{\tau}) \xleftrightarrow{FT} F(\mathbf{u}) e^{i\mathbf{u}\boldsymbol{\tau}^T}. \quad (7)$$

The above equation is consistent with the above definition of the generalized translation, which can be easily verified.

Based on the generalized translation expression, several generalized convolution theorems based on various transformation kernels have been proposed [19,49,57]. However, these all

focused on 1-D signals, and the generalized transformation of a multidimensional signal space remains open.

3. Two types of convolution theorems for the FMT domain

In this section, we present two new classes of convolution theorems for the FMT domain. The expressions of the new convolution theorems in the FMT domain are all expressed as simple product forms of two FMTs and are unaffected by any chip multiplier terms. These solutions are elegant and resemble to classical FT results.

3.1. Convolution theorem of the first type

For the special case of the FMT, consider the weight function $\rho(\mathbf{u})$ to be unity and the kernel to be equal to the kernel of the MFT given by Eq. (2); then, the generalized translation in Eq. (6) can be expressed as follows:

$$\begin{aligned} f(\mathbf{t}\theta\boldsymbol{\tau}) &= \int_{\mathbb{R}^N} \rho(\mathbf{u}) \widehat{f_{\mathbf{M}}}(\mathbf{u}) \mathcal{K}_{\mathbf{M}}(\mathbf{t}, \mathbf{u}) \mathcal{K}_{\mathbf{M}}^*(\mathbf{t}, \mathbf{u}) d\mathbf{u} \\ &= \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} \widehat{f_{\mathbf{M}}}(\mathbf{u}) e^{\pi i(\mathbf{u}DB^{-1}\mathbf{u}^T + \boldsymbol{\tau}B^{-1}A\boldsymbol{\tau}^T) - 2\pi i\boldsymbol{\tau}B^{-1}\mathbf{u}^T} \\ &\quad \times e^{-\pi i(\mathbf{u}DB^{-1}\mathbf{u}^T + \mathbf{t}B^{-1}A\mathbf{t}^T) + 2\pi i\mathbf{t}B^{-1}\mathbf{u}^T} d\mathbf{u} \\ &= \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} \widehat{f_{\mathbf{M}}}(\mathbf{u}) e^{\pi i(\boldsymbol{\tau}B^{-1}A\boldsymbol{\tau}^T - \mathbf{t}B^{-1}A\mathbf{t}^T) + 2\pi i(\mathbf{t} - \boldsymbol{\tau})B^{-1}\mathbf{u}^T} d\mathbf{u}. \end{aligned} \quad (8)$$

The function defined by Eq. (8) is called the $\boldsymbol{\tau}$ -generalized translation $f(\mathbf{t}\theta\boldsymbol{\tau})$ of the signal $f(\mathbf{t})$ based on FMT.

First, we introduce a new concept: a generalized convolution operator $\star^{\mathbf{M}}$, which is related to a symplectic matrix $\mathbf{M}$, a generalized version of [19].

Definition 1. Let $f(\mathbf{t}\theta\boldsymbol{\tau})$ denote the $\boldsymbol{\tau}$ -generalized translation of $f(\mathbf{t})$ in the FMT domain, and let the free metaplectic convolution be denoted by $\star^{\mathbf{M}}$, which can be defined as follows:

$$\begin{aligned} (f \star^{\mathbf{M}} g)(\mathbf{t}) &= \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\mathbf{t}\theta\boldsymbol{\tau}) e^{\pi i[\mathbf{u}(DB^{-1} - B^{-T}D^T)\mathbf{u}^T]} \\ &\quad \times e^{\pi i[\mathbf{t}(B^{-1}A - A^TB^{-T})\mathbf{t}^T] - 2\pi i(\mathbf{t}B^{-1}\mathbf{u}^T - \mathbf{u}B^{-T}\mathbf{t}^T)} d\boldsymbol{\tau}. \end{aligned} \quad (9)$$

Based on the generalized free metaplectic convolution operator $\star^{\mathbf{M}}$, we can derive the generalized convolution theorem associated with FMT as follows:

Theorem 1. Let $z(\mathbf{t}) = (f \star^{\mathbf{M}} g)(\mathbf{t})$, $Z_{\mathbf{M}}$, $F_{\mathbf{M}}$, and $G_{\mathbf{M}}$ denote the FMT of $z(\mathbf{t})$, $f(\mathbf{t})$, and $g(\mathbf{t})$, respectively. Then, we have,

$$Z_{\mathbf{M}}(\mathbf{u}) = F_{\mathbf{M}}(\mathbf{u}) \cdot G_{\mathbf{M}}(\mathbf{u}), \quad (10)$$

and,

$$z(\mathbf{t}) = \mathcal{L}_{\mathbf{M}^{-1}}[F_{\mathbf{M}}(\mathbf{u})G_{\mathbf{M}}(\mathbf{u})](\mathbf{t}). \quad (11)$$

Moreover,

$$\mathcal{L}_{\mathbf{M}}[f(\mathbf{t}) \cdot g(\mathbf{t})](\mathbf{u}) = (F_{\mathbf{M}} \star^{\mathbf{M}} G_{\mathbf{M}})(\mathbf{u}). \quad (12)$$

Proof. From Eq. (8), it follows that,

$$g(\mathbf{t}\theta\boldsymbol{\tau}) = \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} G_{\mathbf{M}}(\mathbf{u}) e^{\pi i(\boldsymbol{\tau} B^{-1} A \boldsymbol{\tau}^T - \mathbf{t} B^{-1} A \mathbf{t}^T) + 2\pi i(\mathbf{t} - \boldsymbol{\tau}) B^{-1} \mathbf{u}^T} d\mathbf{u}. \quad (13)$$

Using Eqs. (9) and (13), we can derive the following result:

$$\begin{aligned} (f \star^{\mathbf{M}} g)(\mathbf{t}) &= \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} G_{\mathbf{M}}(\mathbf{u}) e^{\pi i(\boldsymbol{\tau} B^{-1} A \boldsymbol{\tau}^T - \mathbf{t} B^{-1} A \mathbf{t}^T)} \\ &\quad \times e^{2\pi i(\mathbf{t} - \boldsymbol{\tau}) B^{-1} \mathbf{u}^T} d\mathbf{u} \times e^{\pi i[\mathbf{u}(DB^{-1} - B^{-T} D^T) \mathbf{u}^T]} \\ &\quad \times e^{\pi i[\mathbf{t}(B^{-1} A - A^T B^{-T}) \mathbf{t}^T] - 2\pi i(\mathbf{t} B^{-1} \mathbf{u}^T - \mathbf{u} B^{-T} \mathbf{t}^T)} d\boldsymbol{\tau}. \end{aligned} \quad (14)$$

According to the definition of FMT, we obtain the following:

$$\begin{aligned} Z_{\mathbf{M}}(\mathbf{u}) &= \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} z(\mathbf{t}) e^{\pi i(\mathbf{u} DB^{-1} \mathbf{u}^T + \mathbf{t} B^{-1} A \mathbf{t}^T) - 2\pi i \mathbf{t} B^{-1} \mathbf{u}^T} d\mathbf{t} \\ &= \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} G_{\mathbf{M}}(\mathbf{u}) e^{\pi i(\boldsymbol{\tau} B^{-1} A \boldsymbol{\tau}^T - \mathbf{t} B^{-1} A \mathbf{t}^T)} \\ &\quad \times e^{2\pi i(\mathbf{t} - \boldsymbol{\tau}) B^{-1} \mathbf{u}^T} d\mathbf{u} \times e^{\pi i[\mathbf{u}(DB^{-1} - B^{-T} D^T) \mathbf{u}^T + \mathbf{t}(B^{-1} A - A^T B^{-T}) \mathbf{t}^T]} \\ &\quad \times e^{-2\pi i(\mathbf{t} B^{-1} \mathbf{u}^T - \mathbf{u} B^{-T} \mathbf{t}^T)} d\boldsymbol{\tau} \times e^{\pi i(\mathbf{u} DB^{-1} \mathbf{u}^T + \mathbf{t} B^{-1} A \mathbf{t}^T) - 2\pi i \mathbf{t} B^{-1} \mathbf{u}^T} d\mathbf{t}. \end{aligned} \quad (15)$$

From the inverse formula of the FMT,

$$f(\mathbf{t}) = \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} F_{\mathbf{M}}(\mathbf{u}) e^{-\pi i(\mathbf{u} B^{-T} D^T \mathbf{u}^T + \mathbf{t} A^T B^{-T} \mathbf{t}^T) + 2\pi i \mathbf{u} B^{-T} \mathbf{t}^T} d\mathbf{u}. \quad (16)$$

Hence, using Eqs. (15) and (16),

$$\begin{aligned} Z_{\mathbf{M}}(\mathbf{u}) &= \left\{ \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) e^{\pi i(\mathbf{u} DB^{-1} \mathbf{u}^T + \boldsymbol{\tau} B^{-1} A \boldsymbol{\tau}^T) - 2\pi i \boldsymbol{\tau} B^{-1} \mathbf{u}^T} d\boldsymbol{\tau} \right\} \\ &\quad \times \left\{ \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} G_{\mathbf{M}}(\mathbf{u}) e^{-\pi i(\mathbf{u} B^{-T} D^T \mathbf{u}^T + \mathbf{t} A^T B^{-T} \mathbf{t}^T) + 2\pi i \mathbf{u} B^{-T} \mathbf{t}^T} d\mathbf{u} \right\} \\ &\quad \times \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} e^{\pi i(\mathbf{u} DB^{-1} \mathbf{u}^T + \mathbf{t} B^{-1} A \mathbf{t}^T) - 2\pi i \mathbf{t} B^{-1} \mathbf{u}^T} d\mathbf{t} \\ &= F_{\mathbf{M}}(\mathbf{u}) \left\{ \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} g(\mathbf{t}) e^{\pi i(\mathbf{u} DB^{-1} \mathbf{u}^T + \mathbf{t} B^{-1} A \mathbf{t}^T) - 2\pi i \mathbf{t} B^{-1} \mathbf{u}^T} d\mathbf{t} \right\} \\ &= F_{\mathbf{M}}(\mathbf{u}) \cdot G_{\mathbf{M}}(\mathbf{u}). \end{aligned} \quad (17)$$

Based on Eq. (17) and the reversible property of FMT, we obtain the following equation:

$$z(\mathbf{t}) = \mathcal{L}_{\mathbf{M}^{-1}}[F_{\mathbf{M}}(\mathbf{u}) G_{\mathbf{M}}(\mathbf{u})](\mathbf{t}).$$

The proofs of Eqs. (10) and (11) are complete. The proof of Eq. (12) is as follows:

According to Eq. (5), we obtain:

$$\begin{aligned} \widehat{f_{\mathbf{M}}}(\mathbf{u}\theta\boldsymbol{\tau}) &= \int_{\mathbb{R}^N} \rho(\mathbf{t}) f(\mathbf{t}) \mathcal{K}_{\mathbf{M}^{-1}}(\mathbf{t}, \mathbf{u}) \mathcal{K}_{\mathbf{M}^{-1}}^*(\mathbf{t}, \boldsymbol{\tau}) d\mathbf{t} \\ &= \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} f(\mathbf{t}) e^{-\pi i(\boldsymbol{\tau} B^{-T} D^T \boldsymbol{\tau}^T + \mathbf{t} A^T B^{-T} \mathbf{t}^T) + 2\pi i \boldsymbol{\tau} B^{-T} \mathbf{t}^T} \end{aligned}$$

$$\begin{aligned}
& \times e^{\pi i(\mathbf{u}B^{-T}D^T\mathbf{u}^T + \mathbf{t}A^TB^{-T}\mathbf{t}^T) - 2\pi i\mathbf{u}B^{-T}\mathbf{t}^T} d\mathbf{t} \\
& = \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} f(\mathbf{t}) e^{\pi i(\mathbf{u}B^{-T}D^T\mathbf{u}^T - \mathbf{t}B^{-T}D^T\mathbf{t}^T) + 2\pi i(\mathbf{t}-\mathbf{u})B^{-T}\mathbf{t}^T} d\mathbf{t}.
\end{aligned} \quad (18)$$

From Eq. (18), it follows that:

$$\begin{aligned}
G_{\mathbf{M}}(\mathbf{u}\theta\boldsymbol{\tau}) &= \widehat{g_{\mathbf{M}}}(\mathbf{u}\theta\boldsymbol{\tau}) \\
&= \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} g(\mathbf{t}) e^{\pi i(\mathbf{u}B^{-T}D^T\mathbf{u}^T - \mathbf{t}B^{-T}D^T\mathbf{t}^T) + 2\pi i(\mathbf{t}-\mathbf{u})B^{-T}\mathbf{t}^T} d\mathbf{t}.
\end{aligned} \quad (19)$$

Using Eqs. (9) and (19), we can derive the following equation:

$$\begin{aligned}
& (F_{\mathbf{M}} \star^{\mathbf{M}} G_{\mathbf{M}})(\mathbf{u}) \\
&= \int_{\mathbb{R}^N} F_{\mathbf{M}}(\boldsymbol{\tau}) G_{\mathbf{M}}(\mathbf{u}\theta\boldsymbol{\tau}) e^{\pi i[\mathbf{u}(DB^{-1}-B^{-T}D^T)\mathbf{u}^T + \mathbf{t}(B^{-1}A-A^TB^{-T})\mathbf{t}^T]} \\
&\quad \times e^{-2\pi i(\mathbf{t}B^{-1}\mathbf{u}^T - \mathbf{u}B^{-T}\mathbf{t}^T)} d\boldsymbol{\tau} \\
&= \int_{\mathbb{R}^N} F_{\mathbf{M}}(\boldsymbol{\tau}) \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} g(\mathbf{t}) e^{\pi i(\mathbf{u}B^{-T}D^T\mathbf{u}^T - \mathbf{t}B^{-T}D^T\mathbf{t}^T)} \\
&\quad \times e^{2\pi i(\mathbf{t}-\mathbf{u})B^{-T}\mathbf{t}^T} d\mathbf{t} \times e^{\pi i[\mathbf{u}(DB^{-1}-B^{-T}D^T)\mathbf{u}^T + \mathbf{t}(B^{-1}A-A^TB^{-T})\mathbf{t}^T]} \\
&\quad \times e^{-2\pi i(\mathbf{t}B^{-1}\mathbf{u}^T - \mathbf{u}B^{-T}\mathbf{t}^T)} d\boldsymbol{\tau} \\
&= \left\{ \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} F_{\mathbf{M}}(\boldsymbol{\tau}) e^{-\pi i(\mathbf{t}B^{-T}D^T\mathbf{t}^T + \mathbf{t}A^TB^{-T}\mathbf{t}^T) + 2\pi i\mathbf{t}B^{-T}\mathbf{t}^T} d\boldsymbol{\tau} \right\} \\
&\quad \times \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} g(\mathbf{t}) e^{\pi i(\mathbf{u}DB^{-1}\mathbf{u}^T + \mathbf{t}B^{-1}A\mathbf{t}^T) - 2\pi i\mathbf{t}B^{-1}\mathbf{u}^T} d\mathbf{t} \\
&= \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} f(\mathbf{t}) g(\mathbf{t}) e^{\pi i(\mathbf{u}DB^{-1}\mathbf{u}^T + \mathbf{t}B^{-1}A\mathbf{t}^T) - 2\pi i\mathbf{t}B^{-1}\mathbf{u}^T} d\mathbf{t} \\
&= \mathcal{L}_{\mathbf{M}}[f(\mathbf{t}) \cdot g(\mathbf{t})](\mathbf{u}).
\end{aligned} \quad (20)$$

□

For the matrix $\mathbf{M} = (A, B; C, D)$, we can obtain different convolution theories by choosing the appropriate matrix parameters, as well as the convolution theorems corresponding to all the integral transformations listed in Table 1. The special cases of Theorem 1 in the FT, FRFT, and LCT domains are as follows:

Corollary 1. For $A = D = \mathbf{0}$ and $B = -C = I_N$, Eq. (10) reduces to the convolution theorem for the classical N -D FT $F(\mathbf{u})$ as follows:

$$\int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\mathbf{t} - \boldsymbol{\tau}) e^{2\pi i(\mathbf{u}\mathbf{t}^T - \mathbf{u}\boldsymbol{\tau}^T)} d\boldsymbol{\tau} \xleftrightarrow{FT} F(\mathbf{u}) G(\mathbf{u}). \quad (21)$$

Proof. For $A = D = \mathbf{0}$ and $B = -C = I_N$, Eq. (8) becomes:

$$f(\mathbf{t}\theta\boldsymbol{\tau}) = f(\mathbf{t} - \boldsymbol{\tau}), \quad (22)$$

then, Eq. (9) equals:

$$f(\mathbf{t}) \star^{\mathbf{M}} g(\mathbf{t}) = \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\mathbf{t} - \boldsymbol{\tau}) e^{2\pi i(\mathbf{u}\mathbf{t}^T - \mathbf{u}\boldsymbol{\tau}^T)} d\boldsymbol{\tau}. \quad (23)$$

Hence, we obtain:

$$\int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\mathbf{t} - \boldsymbol{\tau}) e^{2\pi i(\mathbf{u}^T - \mathbf{u}^T \boldsymbol{\tau})} d\boldsymbol{\tau} \xleftrightarrow{FT} F(\mathbf{u}) G(\mathbf{u}). \quad (24)$$

The proof of [Corollary 1](#) is straightforward, simplifying the proof process. $\square$

Corollary 2. For $A = D = \text{diag}(\cos \alpha_1, \dots, \cos \alpha_N)$ and $B = -C = \text{diag}(\sin \alpha_1, \dots, \sin \alpha_N)$, [Eq. \(10\)](#) reduces to the convolution theorem for the N -D FRFT $F_\alpha(\mathbf{u})$ as follows:

$$f(\mathbf{t}) \star^M g(\mathbf{t}) \xleftrightarrow{FRFT} F_\alpha(\mathbf{u}) G_\alpha(\mathbf{u}). \quad (25)$$

Proof. The proof of this property is similar to that of [Theorem 1](#). The result can be obtained using [Eq. \(9\)](#), with $A = D = \text{diag}(\cos \alpha_1, \dots, \cos \alpha_N)$ and $B = -C = \text{diag}(\sin \alpha_1, \dots, \sin \alpha_N)$. $\square$

Corollary 3. For $A = \text{diag}(a_{11}, \dots, a_{NN})$, $B = \text{diag}(b_{11}, \dots, b_{NN})$, $C = \text{diag}(c_{11}, \dots, c_{NN})$ and $D = \text{diag}(d_{11}, \dots, d_{NN})$, [Eq. \(10\)](#) reduces to the generalized convolution theorem for the N -D LCT $F_{(A,B,C,D)}(\mathbf{u})$ as follows:

$$f(\mathbf{t}) \star^M g(\mathbf{t}) \xleftrightarrow{LCT} F_{(A,B,C,D)}(\mathbf{u}) G_{(A,B,C,D)}(\mathbf{u}). \quad (26)$$

Proof. The result can be obtain just by [Eq. \(9\)](#) with $A = \text{diag}(a_{11}, \dots, a_{NN})$, $B = \text{diag}(b_{11}, \dots, b_{NN})$, $C = \text{diag}(c_{11}, \dots, c_{NN})$, and $D = \text{diag}(d_{11}, \dots, d_{NN})$. $\square$

[Theorem 1](#) states that the generalized convolution of two signals in the spatial domain is equivalent to the simple multiplication of their FMTs in the FMT domain. The advantage of this convolution theorem is that it has elegance and simplicity comparable to the classical FT results and is easy to implement, particularly in filter design. However, the disadvantage is that the generalized convolution expression proposed in this study comprises a triple integral form, making it complicated to convert into a single integral form.

3.2. Convolution theorem of the second type

Because the generalized convolution in [Section 3.1](#) is a triple integral, reducing the expression to a single integral form might be challenging. Therefore, we introduce a new convolutional structure represented by a simple integration in the FMT domain that is easy to implement in multiplicative filter designs.

Subsequently, we provide a convolution operator $\star^M$ associated with the FMT.

Definition 2. For any two signals $f(\mathbf{t})$ and $g(\mathbf{t})$, let us define the free metaplectic convolution $\star^M$ as follows:

$$(f \star^M g)(\mathbf{t}) = \frac{\sqrt{2}}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\sqrt{2}\mathbf{t} - \boldsymbol{\tau}) e^{2\pi i\left(\frac{\mathbf{t}}{\sqrt{2}} - \boldsymbol{\tau}\right)^T B^{-1} A \left(\frac{\mathbf{t}}{\sqrt{2}} - \boldsymbol{\tau}\right)} d\boldsymbol{\tau}. \quad (27)$$

A new convolution theorem for FMT can be deduced, as shown in the following theorem.

Theorem 2. Let $z(\mathbf{t}) = (f \star^M g)(\mathbf{t})$, Z_M , F_M , and G_M denote the FMT of $z(\mathbf{t})$, $f(\mathbf{t})$, and $g(\mathbf{t})$, respectively. Hence:

$$Z_M(\mathbf{u}) = F_M\left(\frac{\mathbf{u}}{\sqrt{2}}\right) \cdot G_M\left(\frac{\mathbf{u}}{\sqrt{2}}\right), \quad (28)$$

and,

$$z(\mathbf{t}) = \mathcal{L}_{\mathbf{M}^{-1}} \left[F_{\mathbf{M}} \left(\frac{\mathbf{u}}{\sqrt{2}} \right) \cdot G_{\mathbf{M}} \left(\frac{\mathbf{u}}{\sqrt{2}} \right) \right] (\mathbf{t}). \quad (29)$$

Proof. According to the definition of FMT and Eq. (27), we obtain the following:

$$\begin{aligned} Z_{\mathbf{M}}(\mathbf{u}) &= \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} z(\mathbf{t}) e^{\pi i (\mathbf{u} D B^{-1} \mathbf{u}^T + \mathbf{t} B^{-1} A \mathbf{t}^T) - 2\pi i \mathbf{t} B^{-1} \mathbf{u}^T} d\mathbf{t} \\ &= \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} \left\{ \frac{\sqrt{2}}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\sqrt{2}\mathbf{t} - \boldsymbol{\tau}) e^{2\pi i \left(\frac{\mathbf{t}}{\sqrt{2}} - \boldsymbol{\tau} \right) B^{-1} A \left(\frac{\mathbf{t}}{\sqrt{2}} - \boldsymbol{\tau} \right)^T} d\boldsymbol{\tau} \right\} \\ &\quad \times e^{\pi i (\mathbf{u} D B^{-1} \mathbf{u}^T + \mathbf{t} B^{-1} A \mathbf{t}^T) - 2\pi i \mathbf{t} B^{-1} \mathbf{u}^T} d\mathbf{t} \\ &= \frac{\sqrt{2}}{|\det(B)|} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\sqrt{2}\mathbf{t} - \boldsymbol{\tau}) e^{\pi i (\mathbf{u} D B^{-1} \mathbf{u}^T + \mathbf{t} B^{-1} A \mathbf{t}^T) - 2\pi i \mathbf{t} B^{-1} \mathbf{u}^T} \\ &\quad \times e^{2\pi i \left[\frac{1}{\sqrt{2}} B^{-1} A \left(\frac{\mathbf{t}}{\sqrt{2}} \right)^T - \frac{1}{\sqrt{2}} B^{-1} A \boldsymbol{\tau}^T - \boldsymbol{\tau} B^{-1} A \left(\frac{\mathbf{t}}{\sqrt{2}} \right)^T + \boldsymbol{\tau} B^{-1} A \boldsymbol{\tau}^T \right]} d\boldsymbol{\tau} d\mathbf{t}. \end{aligned} \quad (30)$$

Let $\lambda = \sqrt{2}\mathbf{t}$, Eq. (30) becomes

$$\begin{aligned} Z_{\mathbf{M}}(\mathbf{u}) &= \frac{1}{|\det(B)|} \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\lambda - \boldsymbol{\tau}) e^{\pi i \left[\mathbf{u} D B^{-1} \mathbf{u}^T + \frac{\lambda}{\sqrt{2}} B^{-1} A \left(\frac{\lambda}{\sqrt{2}} \right)^T \right] - 2\pi i \frac{\lambda}{\sqrt{2}} B^{-1} \mathbf{u}^T} \\ &\quad \times e^{2\pi i \left[\frac{1}{2} B^{-1} A \left(\frac{\lambda}{2} \right)^T - \frac{1}{2} B^{-1} A \boldsymbol{\tau}^T - \boldsymbol{\tau} B^{-1} A \left(\frac{\lambda}{2} \right)^T + \boldsymbol{\tau} B^{-1} A \boldsymbol{\tau}^T \right]} d\boldsymbol{\tau} d\lambda \\ &= \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} g(\lambda - \boldsymbol{\tau}) e^{\pi i \frac{\mathbf{u}}{\sqrt{2}} D B^{-1} \left(\frac{\mathbf{u}}{\sqrt{2}} \right)^T + \pi i (\lambda - \boldsymbol{\tau}) B^{-1} A (\lambda - \boldsymbol{\tau})^T - 2\pi i (\lambda - \boldsymbol{\tau}) B^{-1} \left(\frac{\mathbf{u}}{\sqrt{2}} \right)^T} \\ &\quad \times d\lambda \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) e^{\frac{\pi i}{2} \mathbf{u} D B^{-1} \mathbf{u}^T + \pi i \boldsymbol{\tau} B^{-1} A \boldsymbol{\tau}^T - 2\pi i \boldsymbol{\tau} B^{-1} \left(\frac{\mathbf{u}}{\sqrt{2}} \right)^T} d\boldsymbol{\tau} \\ &= G_{\mathbf{M}} \left(\frac{\mathbf{u}}{\sqrt{2}} \right) \frac{1}{|\det(B)|^{\frac{1}{2}}} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) e^{\pi i \frac{\mathbf{u}}{\sqrt{2}} D B^{-1} \left(\frac{\mathbf{u}}{\sqrt{2}} \right)^T + \pi i \boldsymbol{\tau} B^{-1} A \boldsymbol{\tau}^T - 2\pi i \boldsymbol{\tau} B^{-1} \left(\frac{\mathbf{u}}{\sqrt{2}} \right)^T} d\boldsymbol{\tau} \\ &= F_{\mathbf{M}} \left(\frac{\mathbf{u}}{\sqrt{2}} \right) G_{\mathbf{M}} \left(\frac{\mathbf{u}}{\sqrt{2}} \right). \end{aligned} \quad (31)$$

□

Similarly, some typical results regarding the convolution theorem of the second type in the FT, FRFT, and LCT domains can be seen as particular cases of the obtained results.

Corollary 4. For $A = D = \mathbf{0}$ and $B = -C = I_N$, Eq. (28) reduces to the ordinary convolution theorem for the classical N -D FT $F(\mathbf{u})$ as follows:

$$\sqrt{2} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\sqrt{2}\mathbf{t} - \boldsymbol{\tau}) d\boldsymbol{\tau} \xleftrightarrow{FT} F \left(\frac{\mathbf{u}}{\sqrt{2}} \right) G \left(\frac{\mathbf{u}}{\sqrt{2}} \right). \quad (32)$$

Proof. For $A = D = \mathbf{0}$ and $B = -C = I_N$, Eq. (27) becomes:

$$(f *^{\mathbf{M}} g)(\mathbf{t}) = \sqrt{2} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\sqrt{2}\mathbf{t} - \boldsymbol{\tau}) d\boldsymbol{\tau}. \quad (33)$$

The proof of this property is similar to that of [Theorem 2](#); thus,

$$\sqrt{2} \int_{\mathbb{R}^N} f(\boldsymbol{\tau}) g(\sqrt{2}\mathbf{t} - \boldsymbol{\tau}) d\boldsymbol{\tau} \xleftrightarrow{FT} F\left(\frac{\mathbf{u}}{\sqrt{2}}\right) G\left(\frac{\mathbf{u}}{\sqrt{2}}\right). \quad (34)$$

□

Corollary 5. For $A = D = \text{diag}(\cos \alpha_1, \dots, \cos \alpha_N)$ and $B = -C = \text{diag}(\sin \alpha_1, \dots, \sin \alpha_N)$, [Eq. \(28\)](#) reduces to the convolution theorem for the N -D FRFT $F_\alpha(\mathbf{u})$ as follows:

$$(f *^M g)(\mathbf{t}) \xleftrightarrow{FRFT} F_\alpha\left(\frac{\mathbf{u}}{\sqrt{2}}\right) G_\alpha\left(\frac{\mathbf{u}}{\sqrt{2}}\right). \quad (35)$$

Proof. The result can be obtained using [Eq. \(27\)](#), with $A = D = \text{diag}(\cos \alpha_1, \dots, \cos \alpha_N)$ and $B = -C = \text{diag}(\sin \alpha_1, \dots, \sin \alpha_N)$. □

Corollary 6. For $A = \text{diag}(a_{11}, \dots, a_{NN})$, $B = \text{diag}(b_{11}, \dots, b_{NN})$, $C = \text{diag}(c_{11}, \dots, c_{NN})$ and $D = \text{diag}(d_{11}, \dots, d_{NN})$, [Eq. \(28\)](#) reduces to the generalized convolution theorem for the N -D LCT $F_{(A,B,C,D)}(\mathbf{u})$ as follows:

$$(f *^M g)(\mathbf{t}) \xleftrightarrow{LCT} F_{(A,B,C,D)}\left(\frac{\mathbf{u}}{\sqrt{2}}\right) G_{(A,B,C,D)}\left(\frac{\mathbf{u}}{\sqrt{2}}\right). \quad (36)$$

Proof. The result can be obtained using [Eq. \(27\)](#) with $A = \text{diag}(a_{11}, \dots, a_{NN})$, $B = \text{diag}(b_{11}, \dots, b_{NN})$, $C = \text{diag}(c_{11}, \dots, c_{NN})$, and $D = \text{diag}(d_{11}, \dots, d_{NN})$. □

[Theorem 2](#) shows that the proposed solution can be represented by a simple integral and is easy to implement in multiplicative filter designs. However, the method only maintains the elegance and simplicity of classical FT results partially, adding a contraction factor of $\frac{1}{\sqrt{2}}$ in the FMT domain.

The two convolution theorems in this study can be expressed in the FMT domain as the simple product forms of two FMTs, which are unaffected by any chirp multiplier terms. Accordingly, by maintaining elegance and simplicity comparable to the classical FT results, the multiplicative filter design process is presented in [Section 4](#).

4. Filter design in the FMT domain

4.1. Multiplicative filtering associated with FMT

Filter design theory in the field of multidimensional signal analysis is still in its infancy and is an essential research topic in the field of digital signal processing. This section discusses multiplicative filters in the FMT domain and shows that the new convolution structures are easy to implement in the design of filters.

The multiplicative filter model in the FMT domain is illustrated in [Fig. 1](#). If $F_M(\cdot)$ is defined as the FMT spectrum of the received signal $r_{\text{in}}(\mathbf{t})$, and $H_M(\cdot)$ is defined as the frequency response of the filter, then the multiplicative effect in [Fig. 1](#) can be expressed as follows:

$$r_{\text{out}}(\mathbf{t}) = \mathcal{L}_{M^{-1}}\left(F_M[r_{\text{in}}(\mathbf{t})](\tilde{\mathbf{u}}) \cdot H_M(\tilde{\mathbf{u}})\right)(\mathbf{t}), \quad (37)$$

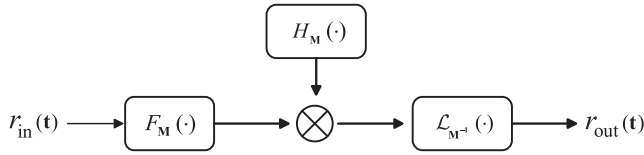


Fig. 1. The multiplicative filter model in the FMT domain.

where $\tilde{\mathbf{u}}$ denotes a generalized variable. The above analysis evidently indicates that the multiplicative filtering process in the FMT domain relies on two steps. That is, first, choosing a suitable filter impulse response and multiplying it by the FMT spectrum of the input signal $r_{\text{in}}(\mathbf{t})$, and second, applying the inverse FMT to the synthesized spectrum to obtain the desired output signal $r_{\text{out}}(\mathbf{t})$. As shown in Fig. 1, by designing $H_{\mathbf{M}}(\tilde{\mathbf{u}})$, we can obtain various multiplicative filter models, including low-pass, high-pass, band-pass, and band-stop filters.

In this section, we provide an example explaining how we can obtain various multiplicative filter models. Suppose that the received signal $r_{\text{in}}(\mathbf{t})$ is composed of the desired signal $f(\mathbf{t})$ and noise $\varepsilon(\mathbf{t})$; that is, $r_{\text{in}}(\mathbf{t}) = f(\mathbf{t}) + \varepsilon(\mathbf{t})$. Moreover, let Ω_1 and Ω_2 be the regions of interest in the FMT domain satisfying $\Omega_1 \subseteq \Omega_2$. Thus, it is reasonable to assume that the FMT spectra of the signals $r_{\text{in}}(\mathbf{t})$, $f(\mathbf{t})$, and $\varepsilon(\mathbf{t})$ have either no or minimal overlap. Consequently, two convolution theorems can be used to design a multiplicative filter in the FMT domain to improve the signal-to-noise ratio, retain the desired signal, and discard noise.

The literature provides two different convolution theorems for FMT; however, the results of their proposed convolution theorems are more complex than ours [33,34]. Note that, in Section 3, we demonstrated that the convolution of two signals in the spatial domain was equivalent to their simple multiplication in the FMT domain. The proposed method maintains the same elegance and simplicity as that of the FT and is easier to implement, particularly in filter design. Based on the two types of convolution theorems introduced in Section 3, here, we discuss the selection method for a multiplicative filter.

- (1) Using the convolution theorem (of the first type) Eq. (10), it can be assumed that the function $G_{\mathbf{M}}(\mathbf{u})$ in Eq. (10) is the transfer function $H_{\mathbf{M}}(\mathbf{u})$ of the multiplicative filter. The function $H_{\mathbf{M}}(\mathbf{u})$ can be selected as follows:

$$G_{\mathbf{M}}(\mathbf{u}) = H_{\mathbf{M}}(\mathbf{u}), \quad (38)$$

such that it is constant over the domain Ω_1 and is zero or decays rapidly outside Ω_1 .

- (2) Using the convolution theorem (of the second type) Eq. (28), assuming that function $G_{\mathbf{M}}(\frac{\mathbf{u}}{\sqrt{2}})$ in Eq. (28) acts as the transfer function of the multiplicative filter, $H_{\mathbf{M}}(\frac{\mathbf{u}}{\sqrt{2}})$ can be designed as follows:

$$G_{\mathbf{M}}\left(\frac{\mathbf{u}}{\sqrt{2}}\right) = H_{\mathbf{M}}\left(\frac{\mathbf{u}}{\sqrt{2}}\right), \quad (39)$$

such that it is constant over domain Ω_2 and is zero or decays rapidly outside Ω_2 .

Fig. 1 shows that the output produces only a part of the $f(\mathbf{t})$ spectrum, which lies approximately in regions Ω_1 or Ω_2 . Here, the complexity of the domains Ω_1 or Ω_2 is not considered. Finally, we can exploit the invertibility of the FMT to recover the desired signal $f(\mathbf{t})$, which is easier to implement than the multiplicative filter designed using the convolution theorems proposed in [33,34].

4.2. Numerical simulations

In this section, we present several simulation experiments, comparing the filter designed in this study with the classical frequency-domain filter [58]. In particular, the experimental results indicated that the filter designed in Section 4.1 was feasible and effective.

To verify the validity of the multiplication filter in the FMT domain, we applied it to two-dimensional image enhancement. In image transmission processing, because the noise is concentrated primarily in the high-frequency part, to remove the noise and improve the image quality, the filter adopted a low-pass filter to suppress the high-frequency component through the low-frequency component, after which the inverse FMT could smooth the image. Based on the convolution theorem, the mathematical expression of the multiplication filter in the FMT domain can be expressed as follows:

$$G_M(u, v) = F_M(u, v)H_M(u, v), \quad (40)$$

where $F_M(u, v)$, $H_M(u, v)$, $G_M(u, v)$ denote the FMT domain of the original image containing noise, transfer function, and the FMT domain of the filtered image output, respectively. Then, we can select the transfer function $H_M(u, v)$ of the multiplicative filter in the FMT domain as follows:

$$H_M(u, v) = e^{-D^2(u, v)/2D_0^2}, \quad (41)$$

where $D(u, v)$ denotes the distance between point (u, v) and the origin, and D_0 denotes the cutoff frequency.

Consider the cameraman image in Fig. 2(a) as an example, as the original gray image used for testing, with an image size of 256×256 . Gaussian noise was added to the original image, and the filter designed in this study was used to smoothen the image and eliminate noise. Additionally, the designed filter in the frequency domain [58] was used as the main comparison object to verify the effectiveness of the proposed method. The structural similarity index (SSIM) and peak signal-to-noise ratio (PSNR) were used to evaluate the image quality after filtering, and the SSIM and PSNR values were calculated directly by using MATLAB SSIM and PSNR commands. The closer the SSIM value is to 1, the better the image quality. The larger the PSNR value, the less distortion.

Case 1: Let $D_0 = 50$, and the free metaplectic matrix:

$$\mathbf{M} = [0.0002, 0, 1, 0; 0, 0.0002, 0, 1; -1, 0, 0, 0; 0, 0, -1, 0, 0].$$

Gaussian noise with a variance of 0.02 was added to the cameraman's image, as shown in Fig. 2(b). The noisy image was filtered based on the classical frequency and FMT domains, the simulation results illustrated in Fig. 2. Fig. 2 clearly shows that the proposed filters are feasible and effective compared with the conventional frequency-domain filtering in [58].

Case 2: Let $D_0 = 50$, and the free metaplectic matrix:

$$\mathbf{M} = [0.0016, 0, -4, 0; 0, 0.0016, 0, -4; 0.25, 0, 0, 0; 0, 0.25, 0, 0].$$

The cameraman's image shown in Fig. 2(a) was perturbed by Gaussian noise with variances of 0.02, 0.05, 0.08, 0.1, 0.15, and 0.2. Table 2 lists the SSIM and PSNR of the output image after the noisy image passes through the filter. The results indicate that the denoising effect of images polluted by different noise levels under the designed filters is stronger than that of the conventional frequency-domain filtering [58].

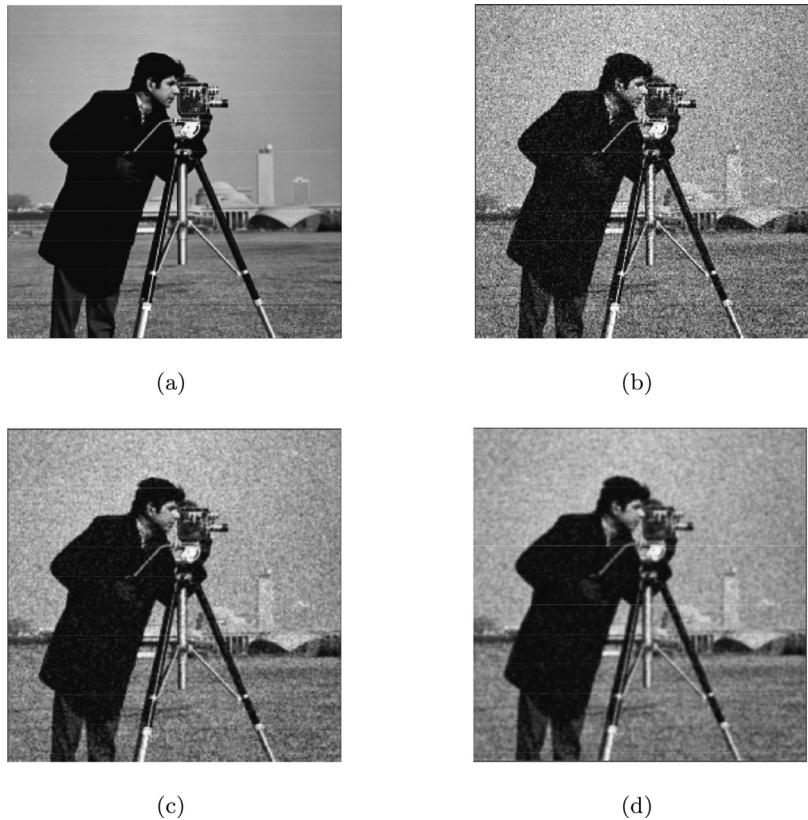


Fig. 2. Image denoising via conventional frequency-domain filtering in [58] and the designed filters. (a) The original image; (b) The noisy image; (c) The output image of the conventional frequency-domain filtering in [58]; (d) The output image of the designed filters.

Table 2
The SSIM and PSNR of [58] and the designed filters under different noise levels.

Noise levels	0.02	0.05	0.08	0.1	0.15	0.2
SSIM (Structure Similarity Index Measure)						
The conventional frequency-domain filtering in [58]	0.5062	0.4920	0.4795	0.4701	0.4549	0.4445
The designed filters	0.6237	0.6064	0.5891	0.5799	0.5573	0.5452
PSNR (Peak Signal-to-Noise Ratio)						
The conventional frequency-domain filtering in [58]	24.0805	22.1218	20.0529	18.7012	15.9124	13.7430
The designed filters	24.6551	22.6262	20.4326	19.0200	16.1262	13.9016

5. Conclusions

In this study, two new convolution theorems in the FMT domain were examined in detail, and their applications were presented. First, based on the generalized translation formulation, we provided a convolution theorem (of the first type) in the FMT domain, which has ele-

gance and simplicity that is comparable to the classical FT results, even though the proposed convolution theorem was in the form of a triple integral and was computationally complex. Second, to further investigate the diversity of convolution theories, we provided the convolution theorem (of the second type) in the FMT domain, which has the advantage that it can be expressed in simple 1-D integrals and is easy to implement in multiplicative filter designs. The two types of convolution theorems proposed in this study were expressed in the FMT domain as simple product forms of two FMTs and were unaffected by any chip product terms. Finally, we examined multiplicative filters in the FMT domain. In future work, we will consider the sampling theorem of FMT in optics and signal processing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Hui Zhao: Writing – original draft, Conceptualization. **Bing-Zhao Li:** Writing – review & editing, Conceptualization.

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References

- [1] I.E. Segal, Foundations of the theory of dynamical systems of infinitely many degrees of freedom, Kgl. Danske Vidensk. Selsk., mat.-fys. Medd. 31 (12) (1959) 1–39.
- [2] D. Shale, Linear symmetries of free boson fields, Trans. Am. Math. Soc. 103 (1) (1962) 149–167.
- [3] N. Wallach, Lie Groups: History, Frontiers and Applications, Symplectic Geometry and Fourier Analysis, vol. 5, Math. Sci. Press, Brookline, MA, 1977.
- [4] V.P. Maslov, M.V. Fedoriuk, Semi-Classical Approximations in Quantum Mechanics, Reidel, Boston, 1981.
- [5] M.A. de Gosson, Symplectic Geometry and Quantum Mechanics, Birkhäuser, Basel, 2006.
- [6] I. Mati, Strongly positive representations of metaplectic groups, J. Algebra. 334 (1) (2011) 255–274.
- [7] M.A. de Gosson, F. Luef, Metaplectic group, symplectic Cayley transform, and fractional Fourier transforms, J. Math. Anal. Appl. 416 (2) (2014) 947–968.
- [8] I. Ciganovic, Reducibility of some generalized principal series of the metaplectic group, Glas. Mat 52 (72) (2017) 257–274.
- [9] H. Ishimoto, Local intertwining relation for metaplectic groups, Compos. Math. 156 (8) (2020) 1560–1594.
- [10] K. Gröchenig, Foundations of Time-Frequency Analysis, Birkhäuser, Boston, 2001.
- [11] R. Howe, E.C. Tan, Non-Abelian Harmonic Analysis, Applications of $SL(2, \mathbb{R})$. Universitext, Springer-Verlag, New York, 1992.
- [12] M.A. de Gosson, Symplectic methods in harmonic analysis and in mathematical physics, in: Pseudo-Differential Operators, Birkhäuser/Springer Basel AG, Basel, 2011.
- [13] V. Kisil, Geometry of Möbius transformations: Elliptic, Parabolic and Hyperbolic Actions of $SL_2(\mathbb{R})$, Imperial College Press, London, 2012.
- [14] F.D. Mari, E.D. Vito, Admissible vectors for mock metaplectic representations, Appl. Comput. Harmon. Anal. 34 (2) (2013) 163–200.

- [15] A.G. Farashahi, Square-integrability of multivariate metaplectic wave-packet representations, *J. Phys. A.* 50 (11) (2017a) 1–22. 115202
- [16] A.G. Farashahi, Square-integrability of metaplectic wave-packet representations on $l^2(\mathbb{R})$, *J. Math. Anal. Appl.* 449 (1) (2017b) 769–792.
- [17] M.A. de Gosson, Symplectic radon transform and the metaplectic representation, *Entropy.* 24 (6) (2022) 1–13.
- [18] B. Deng, T. Ran, Y. Wang, Convolution theorems for the linear canonical transform and their applications, *Sci. China Ser. F-Inf. Sci.* 49 (5) (2006) 592–603.
- [19] D.Y. Wei, Q.W. Ran, Y.M. Li, J. Ma, L.Y. Tan, A convolution and product theorem for the linear canonical transform, *IEEE Signal Process. Lett.* 16 (10) (2009) 853–856.
- [20] J.J. Healy, M.A. Kutay, H.M. Ozaktas, J.T. Sheridan, Linear canonical transforms: theory and applications, in: *Springer Series in Optical Sciences*, vol. 198, Springer, New York, 2016.
- [21] D.F.V. James, G.S. Agarwal, The generalized fresnel transform and its application to optics, *Opt. Commun.* 126 (4–6) (1996) 207–212.
- [22] S.C. Pei, J.J. Ding, Eigenfunctions of fourier and fractional fourier transforms with complex offsets and parameters, *IEEE Trans. Circuits Syst. I.* 54 (7) (2007) 1599–1611.
- [23] Z.W. Li, W.B. Gao, B.Z. Li, The solvability of a class of convolution equations associated with 2D FRFT, *Mathematics.* 8 (11) (2020) 1–12.
- [24] G.B. Folland, *Harmonic Analysis in Phase Space*, Princeton University Press, Princeton, NJ, USA, 1989.
- [25] Z.C. Zhang, New Wigner distribution and ambiguity function based on the generalized translation in the linear canonical transform domain, *Signal Process.* 118 (2016) 51–61.
- [26] Z.C. Zhang, Uncertainty principle for real functions in free metaplectic transformation domains, *J. Fourier Anal. Appl.* 25 (6) (2019) 2899–2922.
- [27] T.K. Garg, W.Z. Lone, F.A. Shah, et al., A convolution-based shearlet transform in free metaplectic domains, *J. Math.* 2021 (2021) 1–23.
- [28] M. Gosson, *Operator Theory: Advances and Applications*, Symplectic Geometry and Quantum Mechanics, Birkhäuser, Basel, Switzerland, 2006.
- [29] J.J. Healy, M.A. Kutay, H.M. Ozaktas, J.T. Sheridan (Eds.), *Linear Canonical Transforms: Theory and Applications*, Springer, New York, 2016, pp. 29–80.
- [30] R. Jing, B. Liu, R. Li, R. Liu, The n-dimensional uncertainty principle for the free metaplectic transformation, *Mathematics.* 8 (10) (2020) 1–15.
- [31] A.Y. Tantary, F.A. Shah, An intertwining of curvelet and linear canonical transforms, *J. Math.* 2020 (2020) 1–14.
- [32] Z.C. Zhang, Uncertainty principle of complex-valued functions in specific free metaplectic transformation domains, *J. Fourier Anal. Appl.* 27 (68) (2021) 1–32.
- [33] F.A. Shah, A.Y. Tantary, Lattice-based multi-channel sampling theorem for linear canonical transform, *Digital Signal Process.* 117 (2021) 1–18.
- [34] F.A. Shah, A.Y. Tantary, Multi-dimensional linear canonical transform with applications to sampling and multiplicative filtering, *Multidimens. Syst. Signal Process.* 33 (2022) 621–650.
- [35] A.Y. Tantary, F.A. Shah, A.I. Zayed, Papoulis’ sampling theorem: revisited, *Appl. Comput. Harmon. Anal.* 64 (2023) 118–142.
- [36] F.A. Shah, H.L. Qadri, W.Z. Lone, Free metaplectic wavelet transform in $l^2(\mathbb{R}^n)$, *Int. J. Wavelets Multiresolut. Inf. Process.* 20 (4) (2022) 1–29.
- [37] A.G. Farashahi, G.S. Chirikjian, Fourier–Bessel series of compactly supported convolutions on disks, *Anal. Appl.* 20 (2) (2022) 171–192.
- [38] A.G. Farashahi, Banach convolution modules of group algebras on covariant functions of characters of normal subgroups, *Bull. Sci. Math.* 185 (2023) 1–19.
- [39] Q. Feng, B.Z. Li, Convolution theorem for fractional cosine-sine transform and its application, *Math. Meth. Appl. Sci.* 40 (10) (2016) 1638–1655.
- [40] L. Durak, Shift-invariance of short-time fourier transform in fractional fourier domains, *J. Frankl. Inst.* 346 (2) (2009) 136–146.
- [41] X.N. Xu, B.Z. Li, X.L. Ma, Instantaneous frequency estimation based on the linear canonical transform, *J. Frankl. Inst.* 349 (10) (2012) 3185–3193.
- [42] R.J. Marks, *Advanced Topics in Shannon Sampling and Interpolation Theory*, Springer-Verlag, 1993.
- [43] Q. Xiang, K.Y. Qin, Convolution, correlation, and sampling theorems for the offset linear canonical transform, *Signal Image Video Process.* 8 (3) (2014) 433–442.

- [44] W.W. Ma, C.J. Xin, D.W. Zhang, Networked continuous-time filtering for quadratically inner-bounded time-delay systems with multirate sampling, *J. Frankl. Inst.* 354 (17) (2017) 7946–7967.
- [45] Y.Y. Han, W.C. Sun, Inversion of the windowed linear canonical transform with Riemann sums, *Math. Meth. Appl. Sci.* 45 (11) (2022) 6717–6738.
- [46] D.Y. Wei, Q.W. Ran, Y. Li, New convolution theorem for the linear canonical transform and its translation invariance property, *Optik*. 123 (16) (2012a) 1478–1481.
- [47] D.Y. Wei, Q.W. Ran, Y.M. Li, A convolution and correlation theorem for the linear canonical transform and its application, *Circuits Syst. Signal Process.* 31 (2012b) 301–312.
- [48] J. Shi, X.P. Liu, N.T. Zhang, Generalized convolution and product theorems associated with linear canonical transform, *Signal Image Video Process.* 8 (5) (2014) 967–974.
- [49] X.Y. Zhi, D.Y. Wei, W. Zhang, A generalized convolution theorem for the special affine fourier transform and its application to filtering, *Optik* 127 (5) (2016) 2613–2616.
- [50] Z.C. Zhang, New convolution and product theorem for the linear canonical transform and its applications, *Optik* 127 (11) (2016) 4894–4902.
- [51] D.Y. Wei, W.W. Yang, Y.M. Li, Lattices sampling and sampling rate conversion of multidimensional bandlimited signals in the linear canonical transform domain, *J. Frankl. Inst.* 356 (13) (2019) 7571–7607.
- [52] D.Y. Wei, Y.M. Li, Reconstruction of multidimensional bandlimited signals from multichannel samples in linear canonical transform domain, *IET Signal Process.* 8 (6) (2014) 647–657.
- [53] M.Y. Bhat, A.H. Dar, Uncertainty principles for the short-time non-separable linear canonical transform, 2022. <https://doi.org/10.48550/arXiv.2204.07048>.
- [54] D.Y. Wei, Y. Shen, Fast numerical computation of two-dimensional non-separable linear canonical transform based on matrix decomposition, *IEEE Trans. Signal Process.* 69 (2021) 5259–5272.
- [55] A.I. Zayed, *Function and Generalized Function Transformations*, CRC, Boca Raton, FL, 1996.
- [56] A.I. Zayed, A product and convolution theorems for the fractional fourier transform, *IEEE Signal Process. Lett.* 4 (1) (1997) 15–17.
- [57] J. Shi, X.J. Sha, et al., Generalized convolution theorem associated with fractional fourier transform, *Wirel. Commun. Mob. Comput.* 14 (13) (2014) 1340–1351.
- [58] M.O. Ling, X. Gao, D. You, Image enhancement of laser welding molten pool feature in frequency domain, in: 2nd International Conference on Electronic and Mechanical Engineering and Information Technology (EMEIT), vol. 23, 2012, pp. 1–4.